



A prospective randomized controlled study of hamstring and bone-free quadriceps tendons autografts in arthroscopic ACL reconstruction

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Abstract

Introduction Anterior cruciate ligament (ACL) reconstruction is one of the most common surgical procedures worldwide. However, the ideal graft source is still debatable. This study compared hamstring (HT) and bone-free quadriceps (QT) tendon autografts in terms of muscle strength and patient-reported outcomes.

Methods Between June 2018 and December 2019, 46 patients were enrolled in the study and randomly assigned to one of two groups for the ACL reconstruction surgery—the HT autograft group or the QT autograft group. An experienced examiner performed preoperative and postoperative examinations. The HUMAC NORM isokinetic machine (Computer Sports Medicine International-CSMI, 2004) was used to assess flexor and extensor thigh muscle strength. Both groups received the same surgical technique and fixation method. All patients were examined after three, six, nine, and 12 months. The final evaluations were performed at the 24-month follow-up. Patients' scores on the International Knee Documentation Committee (IKDC) evaluation form and the Lysholm Knee Scoring Scale were used to evaluate patient-reported outcomes. The two groups were also compared in terms of postoperative complications and activity levels.

Results The difference between preoperative and postoperative knee flexor peak torques (FPT) was lower in the QT group than in the HT group. The average difference between preoperative and postoperative knee extensor peak torque (EPT) was higher in the QT group than in the HT group. As a result, the HT group had a lower postoperative H/Q ratio (flexor strength normalized to body mass divided by extensor strength normalized to body mass) than the QT group ($p < 0.001$). The mean IKDC scores for the HT and QT groups were 90.13 ± 6.99 and 87.12 ± 6.61 , respectively. The scores of the two groups postoperatively were close, and both groups showed significant improvement. The average graft size in the QT group was significantly greater than that in the HT group. Saphenous nerve palsy was the most common complication in the HT group (22%). Thigh hematomas were observed in two patients in the QT group and required urgent intervention (12%).

Conclusion After ACL reconstruction, we found a significant decrease in strength at the donor sites in both techniques, with similar patient-reported measures and stability outcomes. The H/Q strength ratio was higher in the QT group postoperatively, which may have a protective effect on the reconstructed ligament.

Level of Evidence I It is a randomized controlled trial.

Keywords Quadriceps tendon · ACL · Hamstring tendon · Knee injury · Reconstruction

Abbreviations

ANOVA	Analyses of the variance
ACL	Anterior cruciate ligament
ACLR	Anterior cruciate ligament reconstruction
BW	Body weight
Cm	Centimeters
CSMI	Computer Sports Medicine International
EPT	Extensor peak torques
FPT	Flexor peak torques
HT	Hamstring tendon
H/Q	Hamstring/quadriceps

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IKDC	International Knee Documentation Committee
IPBSN	Infrapatellar branch of the saphenous nerve
KOOS	The Knee Injury and Osteoarthritis Outcome Score
Mm	Millimeters
Nm	Newton-meters
PROM	Patient-reported outcome measures
PT	Patellar tendon
QT	Quadriceps tendon
SD	Standard deviation
TC	Thigh circumference

Introduction

The graft choice for ACL reconstruction remains controversial among sports medicine surgeons [1], yet graft source may be one of the decisions that contributes the most to ACL reconstruction outcomes. Though HT and patellar tendon (PT) autografts have proven track records, QT autografts are also gaining popularity and becoming a viable option [2]. Despite widespread use, HT has disadvantages, such as higher re-rupture rates, unpredictable graft sizes, and saphenous nerve palsies [3]. Therefore, there is also a need for alternative graft sources in multi-ligamentous injuries.

QT is gaining popularity due to its advantages over HT, including less donor site morbidity, faster osteointegration, and a lower re-rupture rate [4–8]. Unlike the unpredictable size of HT autografts, QT allows the harvesting of grafts of predictable sizes. Hamstring muscles work as ACL agonists due to their anatomical position, and they compensate for the ACL deficiency in the injured knee and keep the joint stable against valgus forces [9]. In knees with weak hamstrings, there is a higher risk of ACL rupture [10].

The motivations for this research and its importance stem from the research published to date. Four studies have compared QT with PT in ACL surgery [11]. However, publications prospectively comparing bone-free QT with HT autografts are limited. Three retrospective comparisons of QT and HT were published before we started this prospective study [5, 11]. Cavaignac et al. compared thigh muscle strength and instability differences between groups. However, no information was provided about patients' baseline statuses, which may have affected the outcomes [5]. Following this study, several studies compared the use of QT and HT in ACL reconstruction using various surgical techniques and methods. Most concluded that the QT autograft was equal to or better than the HT [7, 8, 12]. Only one study found inferior results with the QT graft [13]. Unfortunately, in most of these publications, the QT autograft was harvested with the patellar bone block (BQT). Altogether, three studies have compared bone-free QT to HT, but only two of these were retrospective designs [14, 15]. Only

Martin-Alguacil et al. designed a prospective study with a two-year follow-up period [16]. However, they used different graft fixation methods for each type of graft, which may have affected stability, strength recovery, and patients' scores on the Patient-Reported Outcome Measures (PROM) scale. Thus, a reliable comparison between HT and bone-free QT autografts has not been conducted to date. Our technique uses bone-free QT autografts, which are easier to harvest and have a lower rate of perioperative complications, such as patella fractures [15], and utilizes the same fixation method in both groups. Our study investigates whether bone-free QT autografts are comparable with HT autografts, using the same fixation method, performed by the same surgeon, with patients rehabilitated by the same center under the same conditions.

This study compared the strength and PROM scores of patients who underwent ACL with a bone-free QT autograft with those of patients who underwent ACL with an HT autograft. We hypothesized that bone-free QT autografts could provide the same outcomes as HT autografts.

Methods

This study was designed as a prospective randomized controlled trial. We received approval from our institution's Ethics Committee (decision no. 1335, dated 07 June 2018). The inclusion criteria were (1) having an isolated primary ACL injury, (2) being willing to join the study, (3) being between 18 and 50 years old, and (4) having no previous surgery in the injured knee. In addition, (1) multi-ligamentous injuries, (2) revision cases, (3) chondral lesions larger than Outerbridge grade 2, and (4) individuals younger than 18 and older than 50 were excluded. All participants provided their written informed consent. Patients were given numbers based on the time of their admittance and were randomized into two groups using the website <https://www.randomizer.org/>. We registered the study to <https://clinicaltrials.gov/>, and the ID is NCT04940169.

Forty patients (20 QT and 20 HT) who met the inclusion criteria were enrolled and operated on within 18 months. Four patients from the HT group and three from the QT group were lost during the follow-up period. Thirty-three patients (three female, 30 male) were analyzed at the end of the study, as shown in the flowchart (Fig. 1). The HT and bone-free QT autografts were clinically compared by patient-reported outcomes, stability tests, and quantitative strength alterations before and after the intervention.

An experienced physician performed preoperative and postoperative clinical evaluations. Patient demographics, trauma mechanism, time-to-surgery, and return to activity were questioned. International Knee Documentation Committee (IKDC) score, Knee Injury and Osteoarthritis

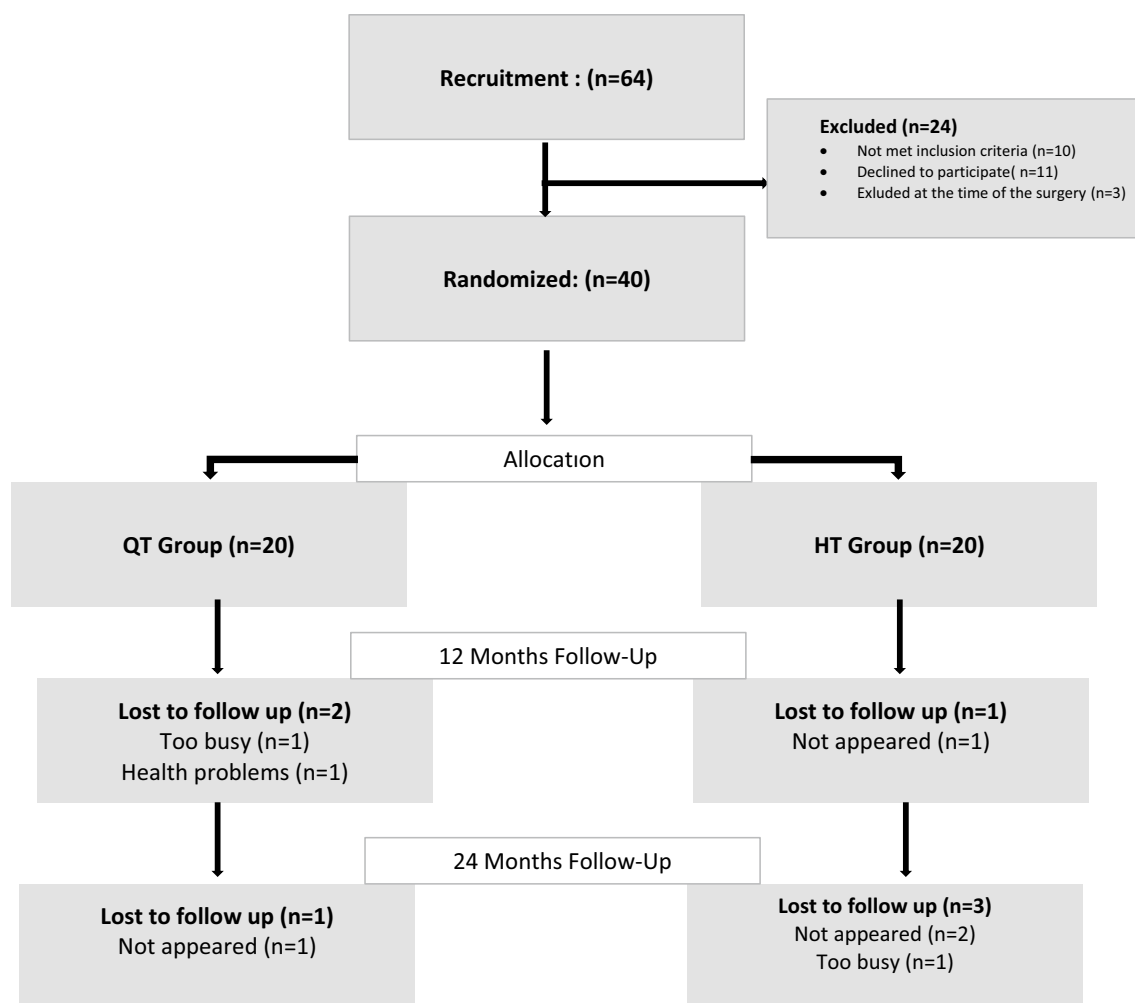


Fig. 1 Flowchart

Outcome Score (KOOS), and Lysholm score were used to follow clinical outcomes. We tested the patients' bilateral hamstring and quadriceps muscle strengths using the Humac CYBEX Norm (CSMI, 2004 Stoughton, MA) isokinetic measuring system a week before and repeated two years after the surgery. Peak muscle torque for knee extension (quadriceps) and flexion (hamstrings) were reported. Due to insufficient strength gaining's and ongoing rehabilitation, we did not need strength measuring in 3, 6, 9, 12 months follow-ups.

Because the strength can alter with size and gender, measurements were normalized by calculating the torque relative to body mass (BW%). The H/Q ratio was calculated by dividing the flexor strength normalized to body mass by the extensor strength normalized to body mass and multiplying by 100. LSI (Limb symmetry index) is calculated by dividing the peak torque power of the injured leg by that of the uninjured leg and multiplying by 100 [17]. In addition, muscle atrophy also was evaluated by measuring the thigh

circumference 15 cm proximal to the proximal pole of the patella since this measurement can represent muscle mass compared to the unaffected leg [18].

Surgical technique

The patients were placed on the operating table in a supine position with both knees at 90° flexion. Administration of spinal anesthesia was preferred. The tourniquet was inflated to 300 mm Hg in all cases. The surgical sites were prepped with betadine solution and covered with disposable drapes. An arthroscopic examination was then performed to confirm that the ACL was ruptured. The same surgeon performed all surgeries.

QT graft harvest

The QT autograft was harvested from a 4 cm longitudinal skin incision starting from the proximal pole of the patella.

A second deeper cut—approximately 7–8 cm long, 1 cm wide, and 6–7 mm thick—was made from the vastus medialis muscle border to the patella using a 15-gauge scalpel in a proximal to distal fashion. The graft included both the rectus femoris and the intermedius tendons. It was incised distally down the patella border and separated from the patellar attachment. After the distal tip was determined, the graft was severed proximally with the help of Metzenbaum scissors and detached from the proximal end [19] (Fig. 2).

HT graft harvest

A 4 cm oblique incision through the skin and subcutaneous tissues was made at the medial side of the proximal tibia. When the pes anserinus was reached, the fascia was opened with an inverted L-shaped cut. Gracilis and semitendinosus were identified, secured with absorbable sutures, and removed using tendon strippers without causing injury to the infrapatellar branch of the saphenous nerve (IPBSN) [20].

Tunnels and fixation

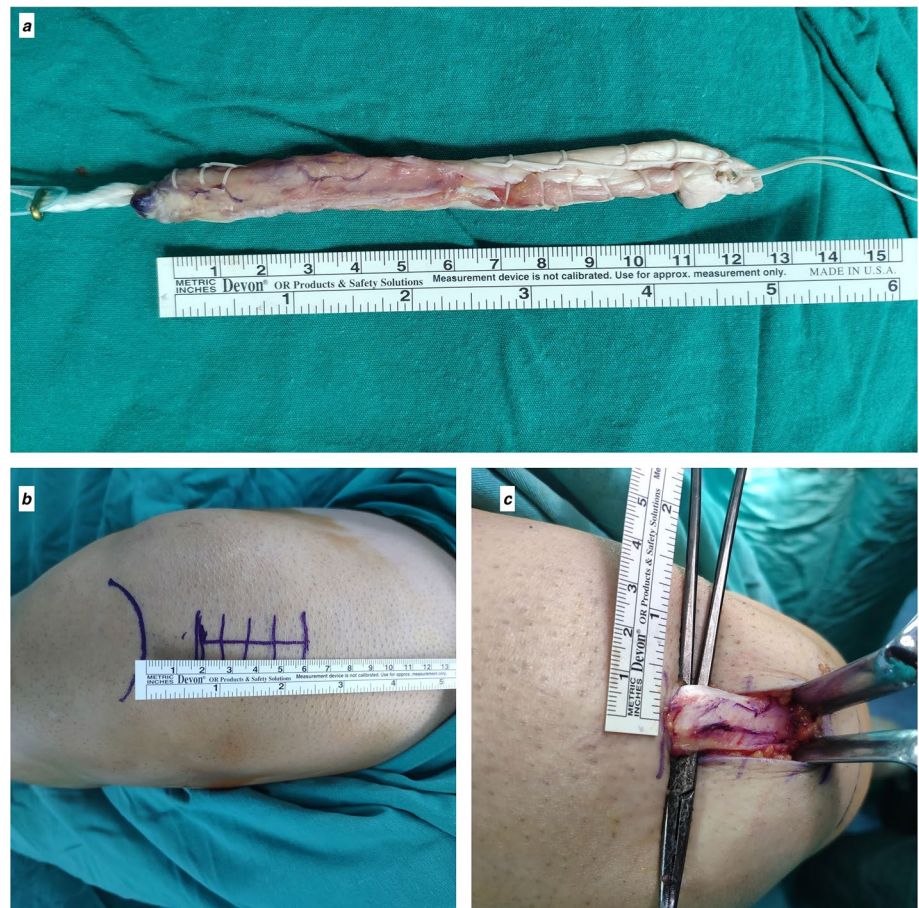
An accessory anteromedial inferior portal was used to locate and reach the femoral tunnel. In the HT group, the tibial

tunnel was drilled using a hamstring graft incision, with the guide adjusted to 55°. In the QT group, an additional skin incision of 1 cm was needed to drill the tibial tunnel. Both femoral and tibial tunnels were created to match the exact size of the grafts. We used the single-bundle technique for ACL reconstruction in all cases in this study. Both grafts were fixed to the femoral tunnel with the help of EndoButton (Smith & Nephew, Andover, USA). The QT graft was attached to the loop using nonabsorbable sutures, and a four-fold hamstring tendon graft was passed through the loop. In both the groups, only the tibial tunnels were fixed with bioabsorbable screws corresponding to the drill diameter and secured with U-shaped staples [21]. Full weight-bearing was allowed first day after surgery. The sutures were removed on postoperative day 15. All patients were directed to an affiliated rehabilitation center to ensure that they followed the same regimen.

Statistical analysis

Statistical analyses were carried out using SPSS version 24.0. The conformity of the variables to the normal distribution was verified using histograms and the Kolmogorov–Smirnov test. Mean, standard deviation, and median

Fig. 2 **a** Loop attached soft-tissue quadriceps tendon with non-absorbable sutures. **b** Incision for the Quadriceps graft harvest. **c** Determining the graft thickness



values were used to present the descriptive analysis. The Pearson chi-square test was used to compare categorical parameters. The Mann–Whitney U test was used to evaluate the non-normally distributed (non-parametric) groups. The Wilcoxon test was used to examine the changes in the measured values. Finally, the McNemar test was used to investigate categorical variables' differences.

The first sample size was calculated using a non-inferiority power analysis for a continuous parameter. The primary endpoint was an H/Q ratio measured using an isokinetic dynamometer in 24 months follow-up. A power analysis was conducted on the primary outcome of the Lysholm test values obtained from the study of Cavaignac et al. [5]. The sample size was estimated to be 16 patients per group, with a power of 80%. The study followed the requirements of the Helsinki Declaration. All participants signed a written informed consent form.

Results

Sixteen patients with HT grafts and seventeen with QT grafts participated in this prospective, randomized controlled study. There were thirty males and three females (all from the HT group), with an average age of 28.27 years. The average time to surgery was 18 months since the injury, with no significant difference between the groups. The most common injury mechanism was knee distortion following landing on an AstroTurf soccer field. The patients underwent independent clinical evaluations and muscle strength tests one week before surgery. The last follow-up was conducted 24 months postoperatively. Noncontact sports injuries were found in 75.8% ($n = 25$) of the patients. Instability ($n = 26$, 78.8%) was the most reported preoperative symptom.

Thirty-three percent (33%) of all patients ($n = 11$) had a meniscal tear. There was no significant difference in meniscus tear frequency between the groups during the preoperative clinical examinations. All participants had

an average graft size of 8.851 mm; the graft size was considerably larger in the QT group than the HT group (9.74 ± 1.55 mm vs. 7.91 ± 0.20 mm, $p < 0.001$) (Table 1). Compared to preoperative status, the flexor peak torque (FPT) and extensor peak torque (EPT) of the HT group were significantly lower than those of the QT group (115.00 ± 22.93 Nm vs. 105.75 ± 22.89 Nm, $p < 0.05$; 210.65 ± 46.77 Nm vs. 156.71 ± 60.26 Nm, $p < 0.05$, respectively) (Table 2).

Despite modest changes in the HT group's extension strength (EPT) and the QT group's flexion strength (FPT), the QT group had a greater H/Q ratio than the HT group at the 24-month follow-up (98.73 ± 26.35 vs. 74.34 ± 12.76 , $p < 0.05$). Because these measurements depend on the patient's physical structure, we adjusted the values for their body weight in the study (%BW), which were more consistent with the results. Thigh circumference (TC) is another basic measurement that indicates postoperative strength reduction. As a result, we computed changes in TC within and between groups. In any case, the findings were not as significant as the isokinetic measurements.

The clinical reflection of graft choice was assessed using IKDC, KOOS, and Lysholm tests, which resulted in average preoperative values of 65.5 ± 9.1 , 69.7 ± 11.3 , and 66.6 ± 11.1 , respectively. Even though neither graft was superior to the other ($p > 0.05$), all patients in both groups showed significant improvement postoperatively (Table 3).

During HT harvest, the IPBSN was frequently injured. Some authors have attributed this to variations in subcutaneous nerve localization [22]. In this study, IPBSN palsy was observed in 25% (4/16) of the patients in the HT group. Two patients in the QT group had hematomas that necessitated further treatment. Hematomas were frequently caused by harvesting a graft longer than 10 cm. Branches of the lateral circumflex artery and lateral geniculate artery have been reported to be injured during the harvest of such a long graft [23]. In addition, one patient in the HT group experienced local implant irritation that necessitated debridement

Table 1 Comparison of the patients' age, height, weight, graft size, and time to operation

	Graft type						<i>P</i> value
	Hamstring			Quadriceps			
	Mean	Sd	Median	Mean	Sd	Median	
Age	28.31	±8.55	27.50	28.06	±6.24	26.00	0.957
Weight (kg)	77.56	±16.93	77.00	77.35	±13.54	78.00	0.885
Height (cm)	173.44	±7.58	173.00	175.65	±5.95	175.00	0.357
Time to surgery (months)	17.41	±18.14	12.00	18.76	±22.77	8.00	0.899
Graft size (mm)	7.91	±.20	8.00	9.74	±1.55	9.00	<0.001

Bold indicates significance is at the $p < 0.05$

Chi-square test

SD standard deviation

Table 2 Comparison of Isokinetic Flexor and Extensor Muscle Strength of HT And QT Groups

	Graft type						<i>p</i> ^a
	Hamstring			Quadriceps			
	Mean	Sd	Median	Mean	Sd	Median	
Injured side thigh circumference pre-op (Cm)	50.00	± 5.80	51.00	49.32	± 5.10	52.00	0.783
Injured side thigh circumference post-op (Cm)	49.19	± 5.46	48.50	48.71	± 5.14	49.00	
<i>p</i> ^b	0.056			0.315			
Injured side EPT pre-op (Nm)	151.06	± 38.11	149.50	159.00	± 26.29	163.00	0.003
Injured Side EPT Post-Op (Nm)	145.06	± 37.37	148.50	119.76	± 41.99	119.00	
<i>p</i> ^b	0.170			0.001			
Injured side EPT pre-op %BW	196.56	± 48.14	194.00	210.65	± 46.77	214.00	0.001
Injured dide EPT post-op %BW	190.19	± 37.24	185.00	156.71	± 60.26	149.00	
<i>p</i> ^b	0.196			0.001			
Injured dide FPT pre-op (Nm)	115.00	± 22.93	118.00	112.35	± 23.43	118.00	0.255
Injured side FPT post-op (Nm)	105.75	± 22.89	113.00	111.88	± 29.33	117.00	
<i>p</i> ^b	0.027			0.586			
Injured side FPT pre-op %BW	151.75	± 32.27	148.00	146.24	± 30.30	141.00	0.305
Injured side FPT post-op %BW	139.19	± 25.73	137.50	145.47	± 36.94	140.00	
<i>p</i> ^b	0.031			0.538			

Bold indicates significance is at the $p < 0.05$

EPT extensor peak torque, FPT flexor peak torque, Pre-Op preoperative, Post-Op postoperative, BW body-weight, SD standard deviation, Nm Newton-meter

^aANOVA for repeated measures

^bWilcoxon test

Table 3 Comparison of examination findings and patient-based tests between preoperative and postoperative HT and QT groups

	Graft type						<i>p</i> ^a
	Hamstring			Quadriceps			
	Mean	SD	Median	Mean	SD	Median	
Lachman pre-op	2.69	± .48	3.00	2.76	± .44	3.00	0.834
Lachman post-op	.00	± .00	.00	.12	± .33	.00	
<i>p</i> ^b	< 0.001			< 0.001			
Anterior drawer pre-op	2.56	± .63	3.00	2.59	± .51	3.00	0.699
Anterior drawer post-op	.00	± .00	.00	.12	± .33	.00	
<i>p</i> ^b	< 0.001			< 0.001			
KOOS pre-op	68.49	± 10.98	69.50	71.47	± 11.57	70.00	0.151
KOOS post-op	92.41	± 3.08	90.50	89.59	± 5.05	90.00	
<i>p</i> ^b	< 0.001			< 0.001			
Lysholm pre-op	65.25	± 11.17	66.50	67.94	± 11.16	69.00	0.531
Lysholm post-op	90.06	± 7.33	91.00	90.00	± 4.26	88.00	
<i>p</i> ^b	< 0.001			< 0.001			
IKDC pre-op	63.66	± 9.84	65.25	66.88	± 8.09	69.00	0.149
IKDC post-op	90.13	± 6.99	92.50	87.12	± 6.61	90.00	
<i>p</i> ^b	< 0.001			< 0.001			

Bold indicates significance is at the $p < 0.05$

^aANOVA for repeated measures

^bWilcoxon test

and removal of the U staple; the patient had no long-term consequences.

Discussion

The most important findings of this study are that the patients in the QT group had better flexion strength. By comparison, those in the HT group had better extensor strength. The clinical tests measuring the stability, flexor strength, and H/Q ratio of patients indicated that the two groups' results were close to each other. As a result, our hypothesis is confirmed.

Our research compared all results within groups and their baseline status as a prospective design, and we followed participants for two years. Although we observed severe strength loss in the graft harvested areas compared to their baseline, the average extensor strength loss was higher than the flexor strength loss. Some researchers have indicated that extensor strength is highly affected, even in HT tendon-harvested ACL surgeries. However, they have also stated that harvesting a graft from the extensor mechanism has a minor effect on flexor strength, while utilizing hamstrings decreases flexor strength massively [24–26]. Furthermore, our results mostly coincide with recent articles [5, 14, 27, 28]. Therefore, if the intent is reserve the flexor strength even after surgery, decision in graft choice before the intervention is valuable.

Fischer et al. compared isokinetic quadriceps and hamstring muscle strength in patients undergoing ACL reconstruction with a QT or HT graft. Within the first year, the QT group had significantly lower extensor strength. By contrast, they declared that a higher H/Q ratio protects the graft in the first months after reconstruction [27]. However, their follow-up period was only 12 months. Despite a large number of patients, the study lacks information about preoperative status, since it had a retrospective design, and they harvested QT tendons with a patellar bone block that could also affect the strength recovery.

Cavaignac et al. and Lee et al. also designed their studies retrospectively and harvested QT grafts with bone block, like Fischer et al. [5, 11]. Therefore, we cannot estimate muscle strength alteration, which may affect the future of the reconstructed graft, from those studies. On the other hand, these studies have proven that harvesting QT tendons has disadvantages, such as extensor strength loss that affects return to sports and even daily ambulation in the early period after intervention.

Recently, Horstmann et al. compared 24 patients with QT to 27 HT, prospectively. They evaluated side-to-side knee laxity with an arthrometer and measured isokinetic

strength, using IKDC and Lysholm scores. Ultimately, both autografts yield good results in primary anterior cruciate ligament reconstruction [28]. However, their fixation method and harvesting quadriceps graft with a bone block make the study different from ours, like other prospective studies [7, 12]. Before that, Martin-Alguacil et al. published a prospective study comparing the HT autograft to bone-free QT in professional soccer players. Eventually, they declared similar functional outcomes and a higher H/Q ratio, which may have a protective impact on the graft [16]. Despite their fixation method differing for each group, using an extracortical loop for HT and bio-screw for the QT, having all the participants be elite athletes, which might affect the strength test results, the study remains unique among all studies comparing HT to QT, and additional studies are needed to strengthen the hypothesis.

We observed that flexor strength recovered faster within the QT group, compared to HT and their baseline values. However, after 24 months of surgery, none of the harvested muscle strength approached their baseline. In a cadaver study, Withrow et al. revealed that increasing hamstring strength reduces relative peak strain on the ACL during knee flexion landing by 70% [29]. Lee et al. observed that flexor strength recovery was better in the QT group as well, with higher knee flexor strength in the first year [11]. A recent meta-analysis also demonstrated that the strength recovery of QT after ACL reconstruction should not be expected to be restored before 24 months [30].

All patients' KOOS, IKDC, and Lysholm scores improved postoperatively compared to the baseline, and the Lachman test scores also improved. After the reconstruction, only one patient in the QT group reported symptoms of instability and pain. Our research yielded results that are like similar to those found in recent publications. Functional outcomes of various graft types have generally been comparable, with no significant difference [31]. Many researchers have demonstrated that QT autograft results are similar to the HT autograft [5, 7, 11, 12, 15, 16, 27]. Lind et al. compared HT and bone-QT grafts in a prospective randomized study. They also revealed that both grafts produced close results in patient-reported outcome measures (PROM), such as KOOS, IKDC, and Tegner activity scores [31]. The only difference between the groups was that the HT group had more saphenous nerve complications, anterior leg pain, and numbness, which could be linked to the injury during the graft harvest. Yang et al. recently reviewed all graft types within 32 trials and 2,962 patients. In addition, the International Knee Documentation Committee (IKDC) subjective and objective scores, the side-to-side difference on knee laxity, the Lachman test, the pivot-shift test, the Lysholm score, the Tegner

score, all recorded complications, and graft re-ruptures were compared. They concluded that the QT autograft is one of the best choices and should be used even over the PT and HT autografts [32].

Considering our results are consistent with the current knowledge, even if it has disadvantages such as extensor strength loss and the need for solid sutures in femoral fixation, using QT may be a better source for the future of the graft and the athlete. However, there is a need for long-term results and larger sample sizes to demonstrate more accurate failure rates.

Some limitations should be noted. The first weakness was the small number of patients, which was difficult to increase, as this study was a randomized controlled trial, and we had a short time frame. However, the overall findings indicate that the number of patients was sufficient to support the hypothesis. The second limitation is having no strength test at 12 months, which could have indicated alterations gradually. Third, 17% of patients were lost during the follow-ups, probably linked to the city's high refugee and asylee population. Finally, obtaining measurements for side-to-side stability differences using an arthrometer would have contributed to our study.

Conclusion

At the last follow-up, patients in the HT group had better extensor strength, while those in the QT group had better flexion strength. In conclusion, given the similar outcomes of the QT and HT grafts, the QT may be a viable option for ACL surgery.

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Author contributions NT Contribution: study design, manuscript preparation; ME Contribution: performed measurements, study design; TG Contribution: critical revision; YO Contribution: manuscript preparation.

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Declarations

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval All procedures performed in studies involving human participants as well as the used registry were by the ethical standards of the institutional review board (IRB) of the Istanbul Training And Research Hospital with decision number 1335, at 07-06-2018 and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all patients included in the study.

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